IT MODULE

Data Processing & Business Analysis

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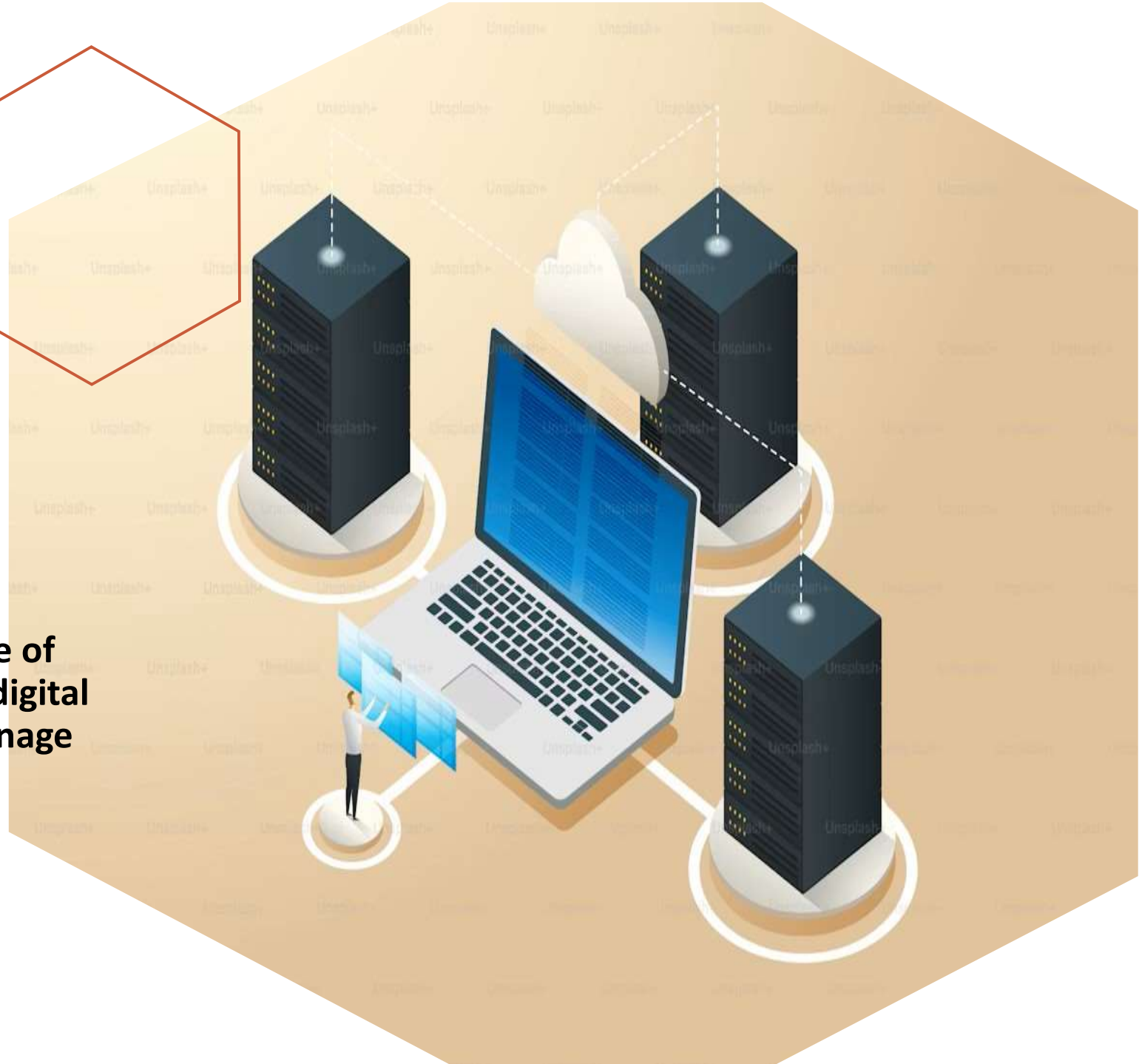
What is Information Technology

Information Technology (IT) refers to the **use of computers, software, networks, and other digital systems** to **store, process, transmit, and manage information**.



IT vs. Computer Science

IT: The use and management of computer systems, networks, and software to store, process, and share information.
Computer Science: The study of computation, algorithms, and the design of computer systems and software.

CORE COMPONENTS OF IT

Hardware

Physical devices (computers, servers, routers, etc.)

Software

Programs and operating systems that run on hardware

Data

The raw facts that are processed into meaningful information

People and Processes

Users, IT professionals, and workflows managing systems

Networking

Connecting systems for data sharing (Internet, LAN, WAN)

Software & Networking

Software Types:

System: OS like Windows or macOS.

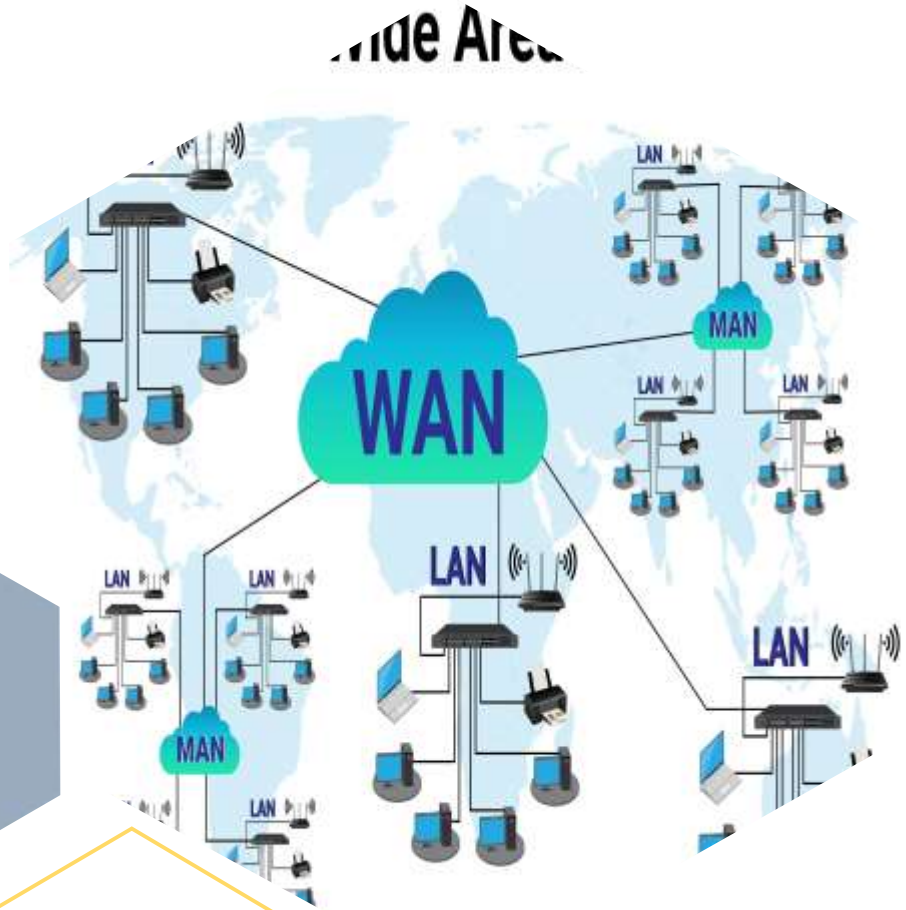
Application: Tools we use (MS Office, Browsers).

Utility: Antivirus and file management tools

Networking 101:

Types: LAN (Local), **WAN** (Wide Area), **PAN** (Personal).

Devices: Routers (directs traffic), Switches (connects devices), and Modems.



Data Management & Cloud Computing

Database Management (DBMS):

Using SQL to organize data for speed and security (e.g., MySQL, Oracle).

Cloud Computing:

- Getting IT services over the internet.
- **Models: SaaS** (Software), **PaaS** (Platform), **IaaS** (Infrastructure).
- **Top Providers:** AWS, Azure, Google Cloud.

Cybersecurity:

- Protecting data from viruses, phishing, and ransomware using firewalls and encryption.

ERP & CRM – Managing the Business

ERP (Enterprise Resource Planning): A "single source of truth" that integrates Finance, HR, and Supply Chain.

Key Feature: Centralized Database.

Examples: SAP, Microsoft Dynamics, Odoo.

ERP vs. CRM:

ERP: Back-office (Internal efficiency).

CRM: Front-office (Customer relationships and sales).

Understanding Data Types

Structured: Neat and tidy. Rows and columns (Excel, SQL).

Semi-Structured: Not a table, but has tags (JSON, XML, Emai

Unstructured: Messy and free-form (Videos, PDFs, Social Me

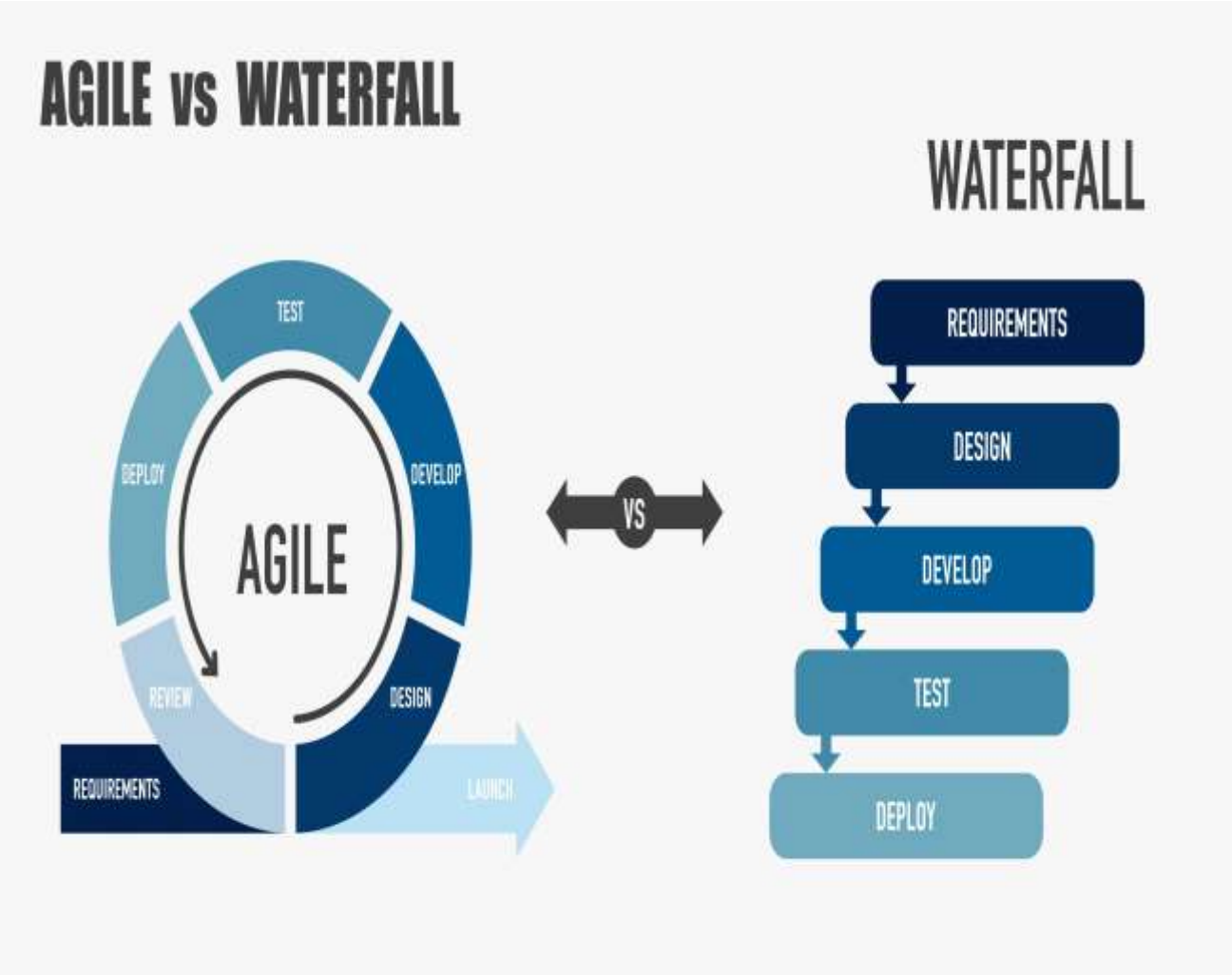
- Sources :

- **Internal:** Your own sales records.
- **Open Data:** Free gov statistics (Weather, Census).
- **Third-Party:** Google Analytics or market research.

Agile vs. Waterfall Methodologies

- **Waterfall:** Step-by-step.
- *Best for:* Clear, unchanging projects (e.g., building a bridge).

- **Agile:** Iterative/Cycles.
- *Best for:* Software development where requirements change fast.
- *Keywords:* Sprints, Scrum, Daily Stand-ups.



Data Governance & Ethics

What is Data Governance?

The framework of rules and processes that ensure data is **accurate, available, and secure**

Key Goals: Improving data quality, managing who has access (Accountability), and ensuring data consistency across the whole company

What are Data Ethics?

The moral principles governing the collection and use of data.

Focus: Transparency, privacy, and fairness (avoiding bias in AI and algorithms).
Key Pillars:
Compliance: Following laws like **GDPR** (Europe) or **CCPA** (California) to protect user privacy.

Stakeholders & Scenario Analysis

Stakeholder Management:

- Identifying and communicating with anyone affected by the project (Customers, Managers, Regulators)

Scenario Analysis:

Preparing for the future by looking at:

- **Best-Case:** Everything goes perfect.
- **Worst-Case:** Major failures or market crashes.
- **Most Likely:** The realistic path.

Conclusion

- IT is no longer just "the computer department"—it is the core of every modern business strategy.
- Successful companies align their **Technology** with their **Business Goals**.